Welcome to Code Crunch!

C7 | CRUNCH LABS: Wire Crunch - Embedded Systems

TUE 1-3PM Weekly | Spring 2025

Attendance

Earn a Certificate of Completion by attending at least 70% of the workshops

linktr.ee/CODE.CRUNCH

Unit #2

C7 | Crunch LABS: Wire Crunch - Embedded Systems

Sensors and Data

What are sensors?

- They are devices that detect and respond to specific physical changes
  - Ex. Light, heat, pressure
- They are used in nearly everything
  - Ex. airbag systems, automatic doors, sensor pads
- The goal of sensor is to detect and respond accordingly and in a correct and safe manor

How Sensors work

- Detection - Sensors will detect a specific physical change
- Conversion - they convert the detected change to electrical signals
- Processing - the signals are processed by the microcontroller to extract the information needed
- Action - the information is relayed and the software designated for it is called upon and performs the action

Reading Sensor values

- We use analog pins to read a sensors value
    - They fluctuate on a scale of 0-1023
- analogRead(pin) will output this value
- The value will tell you what is happening with the sensor
- Use Serial to print out the value that way we can understand it outselves

Moisture sensors

- We want to read the value of the moisture sensor so we will use analogRead(moisturePIN)
- Moisture sensors measure the level of moisture in the soil
  - In arduino it measures the electrical resistance of the soil

Debugging out sensors

- First we want to make sure we are printing out our analog values with analogRead
- If the values are correct, we should include debug statements with Serial.println()
- Ensure your pins are correctly plugged in and declared in the setup
- If you were doing this physically you could use a voltmeter which measures the voltage in the electrical circuits
- Unit test each individual part to make sure functionality is working properly

Debugging out sensors

- First we want to make sure we are printing out our analog values with analogRead
- If the values are correct, we should include debug statements with Serial.println()
- Ensure your pins are correctly plugged in and declared in the setup
- If you were doing this physically you could use a voltmeter which measures the voltage in the electrical circuits
- Unit test each individual part to make sure functionality is working properly

Q&A Session

CODE CRUNCH

Attendance

linktr.ee/CODE.CRUNCH

Join us for the next sessions

Thursdays: 11AM – 12PM & 2PM – 3PM
Friday: 3PM – 4PM
Monday: 12PM – 1PM

RSVP lu.ma/CODECRUNCH

Your feedback helps us improve. Share your thoughts!
Go to our website: go.fiu.edu/codecrunch